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EDUCATION

M.Sc., Human and AI 2025 — 2027

University of Technology Nuremberg (UTN)

Focus: whether AI models actually use the causal structure that language encodes to license inference, or only its surface — studied with causal interpretability.

German language program 2024 — 2025

University of Marburg · intensive German for graduate study

B.A., Philosophy 2020 — 2024

Shenzhen University · College of Humanities

Senior thesis: *On Leibniz's Logical Theory* (advisor: Zang Yong)

Coursework: logic, history of philosophy, analytic philosophy, mathematical logic; electives in cognitive psychology and linguistics. This background shapes my focus on reasons, causality, and what makes an inference licensed.

PAPERS & PLANNED OUTPUTS

Stable Is Not Grounded workshop paper · in submission

Binary certification for whether the licensing structure is load-bearing — the piece written for publication (BlackboxNLP target).

[Project page ↗](#) · [GitHub ↗](#)

Isotrace decision experiment · next

Behavioral path localization via path-encoded fictional names; runs first as a go/no-go on whether it separates genuine multi-hop reasoning from shortcut-copying, and becomes the second paper if it does.

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PROJECTS

Arrowhead personal · pilot complete

A completed mechanistic pilot on a real LLM (Gemma-2-2B). Linear probes, activation steering, and layer-band ablation show the model holds a representation of causal direction — which event is the cause — that is linearly decodable (cross-domain peak ≈ 0.94), causally efficacious under steering, and abstract enough to transfer to fictional, zero-frequency relations, yet redundantly distributed across layers (ablation cannot cleanly remove it). End-to-end interpretability pipeline designed, run, and analyzed on an HPC GPU cluster.

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HanGL — surface Hangul, latent Hanja

collaborative

The structure-versus-surface question run on real LLMs through a natural phenomenon: Korean Sino-Korean words, where Hangul shares no characters with its Hanja origin yet matches it in meaning. A wrong-Hanja intervention shows latent grounding scales with model capability — strong models neither need the cue nor are misled by it; a weak one is both. Pilot complete; scaling up.

[Project page ↗](#) · [GitHub ↗](#)

Latent Control States

collaborative

Does prompt framing — urgency, authority, epistemic positioning — only change an LLM's outputs, or shift a causally relevant latent state? A controlled two-agent data pipeline is built; linear probing and activation patching come next. A collaborative project carried out with, and advised by, a senior researcher at DFKI (the German Research Center for Artificial Intelligence).

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Cross-lingual ProtoBias

collaborative

An applied multimodal stress test: does a vision-language model track semantic content, or language- and culture-specific prototype shortcuts? Extends a forthcoming paper from an established NLP research group with a cross-lingual evaluation, now at its second iteration.

[Project page ↗](#) · [GitHub ↗](#)

MiniCausalLang

personal · workbench

A language-to-causal-graph workbench that makes grammatical structure, the inference it licenses, and interventions on it observable — the shared substrate the certify / localize / quantify methods are built and tested on.

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SKILLS

Consolidating the technical stack my research program needs — controlled linguistic data generation, transformer-based model evaluation, intervention-based causal testing, and reproducible experiment pipelines.

Programming & data

Python (NumPy, pandas, scikit-learn), Git, LaTeX.

ML / NLP tooling

Building working proficiency with PyTorch and HuggingFace Transformers — model inference, dataset pipelines, and LoRA fine-tuning.

Causal & interpretability methods

Controlled do-interventions, causal graphs, and shortcut-severing audits; developing behavioral path-tracing.

Linguistic evaluation

Designing controlled stimuli for morphology, agreement, dependency structure, and word order — drawing on grammar-sensitive work across Chinese, German, English, and Japanese.

LANGUAGES

Mandarin Chinese (native) · German (C1) · English (C1) — spanning isolating, fusional, and analytic types. Learning Japanese, to extend the range with an agglutinative, head-final system — the typological span my linguistic work draws on.

INTERESTS

The causal structure language encodes, and whether AI models use it or only its surface; causal interpretability as the lens. The boundary between behavioral and mechanistic methods. Philosophy of measurement and of language in ML. World-model approaches (LeCun, Pearl) and the limits of pure-LLM paradigms.